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Section 5.1: Data Types

This section will take a look at the different categories of data that can be used in a computer. We call these categories '**Data Types**'.

When we structure data into a spreadsheet or a database we have to tell the computer what data type to use for each piece of data.

Key Concepts of this section:

- # What is **data**?
- # Know what **data types** are and how to correctly identify them
- # Be able to select the **correct data type** for different sets of data

What is Data?

Definition: *"Information that is either digital or analogue and can be manipulated to produce a result."*

Overview of data:

Data can exist in many forms. For example:

- Text (letters/words)
- Numbers (whole numbers, decimals, currency)
- Date



Key Words:

Digital, Analogue, Text, Numbers, Dates

Examples of Text	Examples of Numbers	Examples of Dates
• This is text • THIS IS TEXT	• 123456 • 10.78	• 19th-Aug-2012 • 19th-August-12

- £99.99
- -18
- 19-08-2012
- 19-08-12

Categories of Data:

There are two main categories of data:

- **Digital Data** - Data that is stored and manipulated on a computer.
- **Analogue Data** - Data which is gathered without a computer.

NOTE!

Digital data is often stored in a database. Databases allow for easy access and searching of the data.

Data Types - Overview

The main types of data found in a computer system are:

1. **Boolean** (True/False) - sometimes also called 'logical'
2. **Text/Alphanumeric**
3. **Numeric** (Numbers)
4. **Currency** (Money)
5. **Percentage**
6. **Date/Time**



Key Words:

Boolean, Text, Number, Date, Currency, Percentage

Examples of Data Types

Boolean	➔	Yes/No - True/False - 1/0
Text/Alphanumeric	➔	22 miles
Numeric	➔	999
Currency	➔	£100
Percentage	➔	50%
Date/Time	➔	21-06-2012 - 17:34PM

The different data types are discussed in more detail below:

1

Boolean Data

Boolean data can only have one of two values - true or false.

Booleans are used when there are only two possible responses to a question. Examples of responses that can be stored as a boolean are:

- True / False
- Yes / No
- On / Off
- Male / Female
- 1 or 0

NOTE!

Booleans can also be shown using tick boxes:

- A tick means **ON**
- No tick means **OFF**

For example: "Tick if you agree to the terms of the website"



TRUE
FALSE

2

Text/Alphanumeric

Text Data:

Text data is used for data that is made up of **letters only**.

For example:- a password using the word '**JUPITER**' would be regarded as **text**.

Alphanumeric Data:

Alphanumeric data is used for data that is made up of **letters AND numbers**.

For example:- a password of '**JUPITER123**' would be regarded as **alphanumeric**.

abc

Examples of Text	Examples of Alphanumeric
Dog	I have a dog
Cat	ABC123
Sunshine	SR67JH8
Water	100 miles
Holiday	madeup@gmail.com
House	13 Cairo street Egypt
Peter	50 pounds
Louise	20°C

NOTE!

Other symbols such as @, #, \$, / and spaces are also considered to be alphanumeric.

3

Numeric Data

Numeric (number) data can be in two forms:- **Integer** and **Real**.

Integer Numbers:

- # Integers are made up of **whole numbers** (numbers without decimals).
- # Integers can be both **positive** and **negative** values.

For example:- the number **123** is an integer. **-123** is also an integer.

Real Numbers:

- # Real numbers are those that include **decimals**.
- # Real numbers can be both **positive** (above zero) and **negative** (below zero).

For example:- the number **33.5** is a real number. **-33.5** is also a real number.

123
3.1

Examples of Integers	Uses of Integers
123456	Number of people in a football

Examples of Real Numbers	Uses of Real Numbers
1.87	Exact height of a person

	stadium
13	Age of a person
8	Number of floors in a hotel
-57	Temperature (positive/negative)
-398576	Number of wheels on a car
0	

	in metres
24.99	Price of an item
5.2	Exact distance in miles
-12.37	Exact temperature
-123.9876	
0.45	

4

Currency Numbers

Currency numbers are **real numbers** (decimals) that have been formatted to include **money symbols** (\$, £ etc).

Currency values that use decimals are usually shown to **2 decimal places**.

For example:- **£19.99** is a typical use of the currency value in England.

Currency values can be both **positive** (above zero) or **negative** (below zero)

£99

Examples of Currency Numbers
£100
\$17.99
-£0.89
€1000000

5

Percentage Numbers

Percentage numbers are **real numbers** (decimals) that have been formatted to show **values out of 100**. For example:

- 0.6 would become 60%

- 0.01 would become 1%
- 1.8 would become 180%

Percentage are usually shown with the **percentage symbol (%)**.

Percentage values can be **positive** and **negative**.

8%

Examples of Percentage Numbers

100%

-20%

-12.5%

25.8%

6

Date/Time Data

Date and Time data can be shown in many different ways. The most common ways are shown in the table below:

23/09/78
8:49am

Examples of Dates

dd/mm/yy (21/06/11)

dd/mm/yyyy (21/06/2011)

dd/mmm/yyyy (21/Jun/2011)

Examples of Times

Long time (17:34:23)

Medium time (05:34:PM)

Short time (17:34)

NOTE!

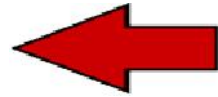
Take care to match the date/time data you are inputting to the style of date/time that your computer is set up to expect.

For example: **American style dates** follow the **mm/dd/yy** format and will cause problems if you try to insert data in the **International style** of **dd/mm/yy**.

Activity!

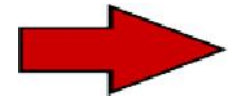
[Data Types - Task](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete both parts of the task sheet.



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